# ANKUR DIXIT

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My research focuses on understanding how climate change and internal variability influence regional hydroclimate, particularly mountain climate systems from the Himalayas to the Rocky Mountains. I also investigate how large-scale atmospheric circulation and local land-surface processes interact to shape regional hydroclimate.

## RESEARCH FOCUS

**01 / CRYOSPHERE**

### Mountain snowpack

Snowpack trends, climate sensitivity and model-resolution effects.

**02 / HYDROCLIMATE**

### Land-atmosphere interaction

Snow–atmosphere/land-atmpsphere coupling in mountain regions.

**02 / EXTREMES**

### Heatwave dynamics/Floods

Circulation, natural variability and forced climate-change signals.

## RESEARCH & PROFESSIONAL APPOINTMENTS

**2023 – present** **Postdoctoral Associate** · Cornell University, Earth & Atmospheric Sciences
Investigating snowpack resilience against climate change over the Upper Colorado River Basin.

**2021 – 2023** **Senior Associate** · PwC India . Part of climate risk and resilience team.

**2014 – 2015** **Junior Research Fellow** · Indian Institute of Remote Sensing (ISRO), Water Resources Department

## EDUCATION

**Ph.D. in Atmospheric Sciences** · Indian Institute of Technology Delhi, India · 2016–2022

## ACCEPTED & SUBMITTED MANUSCRIPTS

**Contrasting dynamical and thermodynamical pathways shaped the 2024 North-Central Indian heatwave.** ACCEPTED
Dixit, A., Najibi N., Mishra N., Dixit J., & Lehner F..

**Resolving snow refugia delays climate change emergence in mountain snowpack.** SUBMITTED
Dixit, A., Musselman K. N., Addor N., Carr S., Gutmann E., Huang L., Minder J. R., Newman A. J., Petach T., Rahimi S., Vano J. A., & Lehner F.

## PEER-REVIEWED PUBLICATIONS

01 **High-resolution models better simulate historical snowpack declines in the Upper Colorado River Basin.**
Dixit, A., Rahimi, S., Huang, L., Musselman, K. N., Vano, J. A., Addor, N., & Lehner, F. (2026). ***Environmental Research Letters.***

02 **Optimizing WRF-Hydro calibration in the Himalayan basin: Precipitation influence and parameter sensitivity analysis.**
Dixit, A., Sahany, S., Mishra, S. K., & Lehner, F. (2025). ***Journal of Hydrometeorology.***

03 **Modeling the climate change impact on hydroclimate fluxes over the Beas basin using a high-resolution glacier-atmosphere-hydrology coupled setup.**
Dixit, A., Sahany, S., & Mishra, S. K. (2023). ***Journal of Hydrology, 627, 130219.***

04 **Seasonal dependent suitability of physical parameterizations to simulate precipitation over the Himalayan headwater.**
Dixit, A., Sahany, S., Mishra, S. K., & Mesquita, M. D. (2023). ***Scientific Reports, 13(1), 4756.***

05 **Role of changing land use and land cover (LULC) on the 2018 megafloods over Kerala, India.**
Dixit, A., Sahany, S., Rajagopalan, B., & Choubey, S. (2022). ***Climate Research, 89, 1–14.***

06 **Glacial changes over the Himalayan Beas basin under global warming.**
Dixit, A., Sahany, S., & Kulkarni, A. V. (2021). ***Journal of Environmental Management, 295, 113101.***

## SERVICE & SCIENTIFIC CONTRIBUTION

**AGU session co-convener** NH31F: Multihazard Early Warning System in the Himalayan Region: From Early Warning to Early Action (2024).

**Journal peer review** Reviewer for scholarly journals.

**Technical report** U.S. Department of Energy (2024), Understanding Decision-Relevant Regional Climate Data Products: Workshop Report (DOE/SC-2021).